Activity 10.3: Stability, Time Stepping, and Information Transfer

Geophysics — Numerical Heat Flow

Purpose

To develop physical intuition for numerical stability in explicit finite difference schemes.

Stability Criterion

For a one-dimensional explicit scheme, stability requires:

$$\Delta t < \frac{\Delta z^2}{2\kappa}.$$

Tasks

1. Explain why this condition depends on thermal diffusivity.
2. Describe what happens physically when Δt is too large.
3. Interpret the stability criterion as a limit on how fast thermal information can propagate between nodes.
4. Use the analogy of a speed limit: why can thermal information not "jump" over nodes in a single time step?

Geological Interpretation

- Why are sedimentary basins numerically easier to model than crystalline rocks?
- Why do high diffusivity materials require smaller time steps?

Key Insight

Numerical instability reflects violation of physical causality, not just poor coding practice.